

External Oracles Accelerate Cultural Codex Circulation Without Rewriting Structural Grammar

Target: PNAS

Status: v8 — reviewer response revision. 2026-06-15.

Figures: Fig. 0 (paired vignette), Figs. 1–5 (main), Figs. S1–S3 (SI).

v8 changes: significance expanded; Codex defined at first use; §2.5 title de-literary-ized; feature standardization disclosed (§4.3); half-life definition + threshold sensitivity added (§4.4); causal-identification paragraph added to Discussion; “general property” softened; selective convergence flagged for SI relocation in response letter.

Significance Statement

Using 95,000 recorded Go games spanning four centuries, we show how an external AI system changed not what professional players could play, but how quickly new opening patterns were discovered and stabilized in a centuries-old cultural record. These findings provide a quantitative framework for understanding how external oracles reshape cultural evolution in domains ranging from art to scientific discovery.

Abstract

The advent of superhuman AI in Go has been widely described as a rupture in traditional playing knowledge. Using 95,000 recorded games from the CWI archive (1600–2026), we factorize each move into separable game-phase and spatial-relation coordinates, measuring decade-level structural drift across four centuries. The AI-era drift magnitude (0.013) lies within the pre-2016 historical envelope (mean 0.015; empirical 95% range [0.004, 0.034]). The largest drift—1910s to 1920s (0.034)—was an institutional revolution (Oteai professional ranking); the AI era is an oracle-speed revolution. Among qualifying recurrent opening patterns, observed adoption lag shortened from 6.3 years (pre-1980) to 1.5 years (post-2016), subject to right-censoring in the post-2016 window. Observed reuse density rose from 0.35 to 0.78. These findings are not artifacts of archive composition: Japan-only replication preserves the drift magnitude (0.0159 vs 0.0160). Regional event-stream dynamics reveal a three-phase response—shock convergence, divergent absorption, partial oracle re-stabilization—consistent with the held-out Codex operation trajectory (Codex: externalized knowledge repository; Human Codex exceeds Frequency in 22 of 22 years; sign test $p < 0.001$). Our results establish that external oracles reshape cultural evolution not by rewriting structural grammar but by accelerating the discovery, transmission, and persistence of new patterns within existing structural constraints.

1. Introduction

In March 2016, AlphaGo defeated Lee Sedol 4–1 in a five-game match watched by a global audience. Professional Go players rapidly adopted AI-influenced strategies, leading to widespread claims that AI had “rewritten Go.” But what exactly changed: the structural grammar of move choice, or the speed at which new moves are discovered and adopted?

Here, *structural grammar* refers to stable constraints linking game phase and spatial move-choice features, operationalized by chroma-rhythm coupling and rhythm-centroid drift across eras. These two possibilities—structural rewriting versus circulation acceleration—are confounded in qualitative accounts. Disentangling them requires (a) a structural encoding that separates game-phase from spatial-choice, (b) a century-scale baseline against which to measure deviation, and (c) a method for tracking pattern adoption speed independent of pattern content. We address all three.

1.1 Case Study: Two AlphaGo Encounters as Microcosm

To motivate the framework, we analyze two landmark human-AI encounters at the single-game level. Fig. 0 shows per-move retrospective KataGo policy alignment and winrate trajectories.

Lee Sedol (Game 4, 2016). Before his celebrated move 78, Lee’s moves were rarely among KataGo’s top recommendations (mean zero-indexed policy rank = 20; 25% in top-1; estimated winrate 1.6%). The divine move itself was in KataGo’s top-3—the third-ranked option by policy order (zero-indexed rank 2)—and the engine’s winrate assessment did not shift.

After move 78, Lee’s choices aligned sharply with KataGo (mean rank = 4; 39% top-1; 83% top-3), and his winrate rose from 1.4% to 99.6%. His post-78 play systematically differed from his play in games he lost to AlphaGo (adj_opp Cohen’s $d = +0.19$; adj_own $d = +0.32$), and AlphaGo’s subsequent moves became more spatially displaced (dist_last $d = +0.55$). Lee appears to have entered an AI-aligned regime through a phase transition.

Ke Jie (Game 1, 2017 Future of Go Summit). One year later, Ke Jie began his game already in an AI-aligned regime (mean zero-indexed rank = 1.4; 61% top-1; 83% top-3). His alignment did not shift—it was high from the start. He nonetheless lost, with his estimated winrate declining steadily from near-even toward zero.

This paired contrast illustrates that retrospective AI alignment is not equivalent to victory. Lee entered an AI-aligned regime through a phase transition; Ke Jie was already there but faced a stronger opponent. Both cases show, in microcosm, how external oracles restructure move-choice regimes without guaranteeing outcomes. The claim of this paper is not about individual moves or games—it is that this pattern characterizes the structural response of professional Go to external AI oracles at scale.

2. Results

2.1 AI-era drift lies within the historical envelope

We computed decade-to-decade structural drift across four centuries using a five-dimensional spatial-relation vector per move (adj_opp, adj_own, dens_delta, dist_last, is_corner_open). The game-phase coordinate captures move index, board occupancy, and phase bin; the spatial-relation coordinate captures distances, adjacencies, densities, and corner proximity to existing stones. Full encoding details are provided in Methods.

The pre-2016 decade-to-decade drift distribution ($N = 15$ transitions) has mean 0.0146 and standard deviation 0.0098. The observed historical range is [0.004, 0.034]. The AI-era transition (2010s to 2020s) has $|drift| = 0.0130$, with Z -score -0.16 relative to the pre-2016 distribution. The AI-era drift lies within the historical envelope (Fig. 1).

The largest drift in our record—1910s to 1920s ($|drift| = 0.034$, $2.6\times$ the AI-era magnitude)—corresponds to the institutionalization of professional Go through the Oteai ranking system and the founding of the Nihon Ki-in (1924). By this structural metric, AI is not the most disruptive event in the CWI record.

None of the five rhythm dimensions individually exceeded the pre-2016 historical range (all $|Z| < 1.96$; is_corner_open showed the largest displacement at $Z = +1.42$).

2.2 Chroma-rhythm coupling remained stable

Cross-harm coupling between game-phase and spatial-relation coordinates was measured across 44,368 post-2000 games. No global shift was detected across AI eras (Cohen’s d , E0 vs E3 = -0.029). The observed AI-era change is best characterized as an operating-point shift: rhythm features moved (Section 2.1) within a stable chroma-rhythm coupling structure.

2.3 Selective convergence toward AI policy

We compared human move choices with KataGo policy recommendations across 200 sampled positions (50 per era, ply 30). The overall human-KataGo rhythm gap did not change across eras (Cohen’s d , E0 vs E3 = -0.008). Feature-level analysis suggested selective convergence on opponent-proximal dimensions (adj_opp, adj_own, dens_delta) but not on pacing or corner strategy. We treat this result as supplementary: it is consistent with the main findings but is based on a limited sample. KataGo analysis used *kata1-b28c512nbt-s13255194368-d5935380940*, maxVisits = 100, Japanese rules, komi 6.5. Full details in Supplementary Information (Fig. S1).

2.4 Pattern adoption accelerated; reuse density increased

We tracked 74,501 unique opening patterns (first 15 moves), of which 546 met the qualifying threshold of ≥ 3 years with ≥ 5 total uses. Per-era metrics are reported in Table 1.

Table 1. Pattern adoption and reuse by era of first appearance.

Era	N patterns	Observed adoption lag (yr)	Observed reuse density	Half-life (yr)
pre-1980	190	6.3	0.35	1.1
1980–1999	154	5.7	0.37	1.1
2000–2015	87	2.6	0.67	1.0
2016+	115	1.5	0.78	1.1

Adoption lag is measured as years from first observed appearance to first year with ≥ 3 unique players, among qualifying recurrent patterns. These metrics are conditional on the qualifying filter and on the observed follow-up window; post-2016 patterns have shorter available follow-up (maximum 9 years vs 50+ years for pre-1980 patterns). Fixed-horizon adoption analysis supports the conclusion that the observed acceleration is not solely a right-censoring artifact: the probability of reaching ≥ 3 unique players within 5 years rose from 0.44 (pre-1980) to 0.91 (2016+) (Fig. 2). Bootstrap resampling of the adoption lag ratio (pre-1980 / 2016+) yields $4.2\times$ (95% CI $[2.9\times, 5.9\times]$; 5,000 resamples). Reuse density increased $2.3\times$ (95% CI $[2.1\times, 2.5\times]$). Pattern half-life remained approximately 1.1 years across all eras.

Long-term adoption acceleration is robust to prefix length: prefix 10 shows $9.0 \rightarrow 1.3$ yr, prefix 15 shows $6.3 \rightarrow 1.5$ yr, prefix 20 shows $4.1 \rightarrow 1.4$ yr. The additional post-2016 reduction is strongest for prefix lengths 10 and 15, but not detectable at prefix length 20, where both 2000–2015 and 2016+ show 1.4-year observed lag (Supplementary Information, Fig. S2).

2.5 Archive composition bias and its control

The CWI archive is itself a product of Codex circulation. Table 2 reports true game counts by decade and event stream.

Table 2. CWI archive composition by decade.

Decade	JPN-coded	KOR-coded	CHN-coded	INTL/Other	Total
1950s	1,668	0	0	200	1,868
1980s	9,640	1	0	106	9,747
2000s	9,198	1,078	4	831	11,111
2020s	12,175	1,642	207	523	14,547

To verify that our structural findings are not artifacts of this composition shift, we replicated the decade drift analysis on a Japan-only subset—the only continuous archival record across all decades. The Japan-only drift magnitude (0.0159) was nearly identical to the full-archive value (0.0160). The drift signal is in the moves, not in the archive composition.

2.6 Two revolutions, two mechanisms

The 1910s→1920s structural drift (0.034) was decomposed by splitting games into overlap players (20 players appearing in both decades) and non-overlap players. Overlap players accounted for 85% of the total drift magnitude (0.0291 of 0.0342), indicating a true playing-style shift rather than roster turnover. This drift accompanied the institutionalization of professional Go (Oteai ranking, Nihon Ki-in founding). The 2010s→2020s drift (0.013) occurred within a stable institutional framework and at average historical magnitude, yet the AI era produced a large increase in pattern circulation speed (Section 2.4) that the institutional revolution did not. The two revolutions operated through different mechanisms: institutional reorganization versus oracle-speed circulation (Fig. 3).

2.7 Regional event-stream dynamics echo the Codex destabilization-recovery pattern

Games were classified into event streams by tournament-name keyword matching into Japan-coded, Korea-coded, and China-coded subsets. Event-stream labels reflect tournament provenance and archival pathway, not player nationality. Inter-stream five-feature spatial-relation distance reveals a three-phase response (Fig. 4):

- Pre-AI (1990–2015): 0.0144 [95% CI 0.0109, 0.0361]

- AlphaGo shock (2016–2017): 0.0066 [95% CI 0.0037, 0.0151]
- AI diffusion (2018–2021): 0.0167 [95% CI 0.0109, 0.0323]
- Oracle era (2022–2025): 0.0102 [95% CI 0.0056, 0.0212]

The Korea-coded event-stream fighting index ($\text{adj_opp} - \text{adj_own}$) peaked during AI diffusion (2018–2021: +0.038), above its pre-AI level (2008–2015: +0.027). AI initially amplified stream-specific signatures before the oracle era pulled them toward the full-archive centroid. This three-phase pattern is consistent with the Codex operation trajectory observed in held-out prediction: Human Codex exceeds the Frequency baseline in all 22 held-out years (sign test $p < 0.001$), with era-level mean Δ values showing a peak at E1, a dip at E2, and recovery at E3 (bootstrap 95% CIs all above zero). Leave-one-out control confirms the Korea-coded stream fighting index is not driven by any single top player ($\Delta < 0.001$ per player; Supplementary Information, Fig. S3).

3. Discussion

Our findings resolve an apparent paradox. Qualitatively, players and commentators report that AI “changed everything.” Quantitatively, the move-level structural drift of the AI era lies within the historical envelope. The resolution is that what changed was not move structure but Codex circulation speed.

This distinction matters beyond Go. It raises the possibility that external oracles in cultural evolution accelerate the discovery–transmission–persistence cycle without necessarily rewriting structural grammar. The grammar—the coupling between game-phase position and spatial-choice—is constrained by the geometry of the board and the deep structure of the game. An oracle can reduce search costs, enabling faster pattern discovery and stronger reuse once a pattern is validated.

The three-phase regional dynamics further suggest that oracle adoption follows a characteristic destabilization-recovery trajectory: initial convergence under shock, followed by asymmetric absorption where pre-existing signatures are temporarily amplified, culminating in partial re-stabilization at permanently higher circulation speed.

A limitation of causal interpretation should be noted. The AI era coincided with other global changes—widespread internet access, the rise of online Go platforms, and increased tournament prize funds—which may have independently accelerated pattern circulation. Our fixed-horizon and Japan-only analyses partially address confounding from data composition, but they do not isolate the causal effect of AI from these co-occurring trends. The acceleration we observe should be understood as characterizing the AI era broadly, not as a causal estimate of AI’s unique contribution.

Several additional limitations apply. First, the CWI archive is Japan-dominant; while Japan-only replication validates the main drift signal, Korea-coded and China-coded event streams remain underpowered relative to the Japan-coded stream. Second, our pattern metrics are conditional on qualifying recurrent patterns and on the observed follow-up window, though fixed-horizon analysis confirms the acceleration is not solely a censoring artifact. Third, the KataGo analysis is based on a small sample (200 positions) and serves as supplementary confirmation. Fourth, event-stream labels reflect tournament and archival provenance rather than player nationality; the “fighting index” refers to the Korea-coded event stream and should not be interpreted as a national style claim without player-level nationality metadata.

4. Methods

4.1 Data

CWI Professional Go Archive: 95,076 SGF records spanning 1600–2026, with five sparse pre-1600 records excluded from the main century-scale analysis [7]. The archive is predominantly Japanese professional and institutional records; Korean and Chinese games enter substantially from the 1990s onward.

4.2 Move encoding

Each move is factorized into two separable coordinate sets. **Game-phase** (chroma): move index, normalized move index (by fixed horizon 300), board occupancy fraction, occupancy bin, and five phase bins (opening, early-mid, mid, late-mid, endgame). **Spatial-relation** (rhythm): distance to last move, distance to nearest stone (any, own, opponent), own/opponent local stone density within radius 3, adjacent own/opponent stone count, edge distance, corner distance, corner-openness

indicator, and own-minus-opponent density difference. Key paper analyses use five rhythm features (adj_opp, adj_own, dens_delta, dist_last, is_corner_open); the full 9 + 12 vectors are used for cross-harm coupling.

4.3 Decade drift

Games are grouped by decade. Per-decade rhythm centroids are computed for the five key features (each bounded to [0, 1] by construction; adj_opp and adj_own are neighbor-count fractions with max 1.0, dist_last is normalized by board diameter, is_corner_open is binary, and dens_delta is a normalized count difference bounded to approximately [-1, 1]). All features share comparable scale, and the Euclidean distance is computed on their raw values without further standardization. The pre-2016 envelope is defined by all decade-pair drifts where the later decade ends ≤ 2010 ($N = 15$ decade transitions).

4.4 Pattern analysis

Opening patterns are defined as the first 15-move prefix string. Qualifying threshold: ≥ 3 years and ≥ 5 total uses across all years, yielding 546 patterns from 74,501 unique prefixes. Adoption lag: years from first observed year to first year with ≥ 3 unique players, among qualifying recurrent patterns. Reuse density: fraction of years between first and last observed year with active usage. Half-life: years from peak-usage year to the first subsequent year with usage $\leq 50\%$ of peak, observed within each pattern's tracked window. Fixed-horizon adoption: probability of reaching ≥ 3 unique players within a fixed follow-up window (1, 2, 3, or 5 years). Bootstrap CIs use 5,000 per-pattern resamples within each era. Threshold sensitivity: under alternative thresholds (≥ 2 years, ≥ 3 uses), the acceleration pattern remains directionally consistent but with increased sample-size variation.

4.5 Event-stream classification

Tournament-name keyword matching. Japan-coded: Meijin, Kisei, Honinbo, Oza, Tengen, Gosei, Judan, NHK, Oteai, and related terms. Korea-coded: Samsung, LG, BC Card, GS Caltex, and related terms. China-coded: Mlily, Bailing, Chinese City League, Mingren, Tianyuan, and related terms. Default classification is Japan-coded (the archive is Japan-origin). The classifier was audited by manual inspection of the top 50 events per category (see Supplementary Information for classification rules and coverage).

4.6 Codex operation

Human Codex: per-position next-move frequency from all training years. Baselines: global move frequency (Frequency), last-three-years per-position frequency (Recency). Held-out protocol: train on years $< t$, evaluate on year t , repeated for each test year 2005–2026. Statistical validation: sign test for Human Codex exceeding Frequency (22/22 years, $p < 0.001$); bootstrap CI (5,000 resamples) on era-level Δ values.

4.7 KataGo analysis

KataGo v1.16.4 OpenCL backend, network *kata1-b28c512nbt-s13255194368-d5935380940*. For the cross-era comparison: 50 positions per era, sampled at ply 30, maxVisits = 100, Japanese rules, komi 6.5. For the paired vignette (Fig. 0): every second move of the human player, maxVisits = 100, same rules and komi. Policy top-1 compared to the actual move via rhythm feature distance. All KataGo results are treated as retrospective policy proxies and reported as supplementary or case-study material.

4.8 Robustness checks

Prefix length: adoption lag acceleration replicated at prefix lengths 10, 15, and 20 (see Supplementary Information, Fig. S2). Japan-only: decade drift analysis replicated on the Japan-only subset. Event-stream top-player leave-one-out: fighting index stable under removal of any individual top player ($\Delta < 0.001$ per player; Fig. S3). Bootstrap CIs: reported for all key ratios using 5,000 per-sample resamples.

5. Data Availability

Analysis scripts and derived aggregate result tables are available at a public repository to be deposited upon acceptance. Source SGF records are available from the CWI Professional Go Archive (<https://homepages.cwi.nl/~aeb/go/games/>) subject to the archive’s terms. Parsed intermediate data and per-game feature files supporting the main figures are available from the authors upon request.

6. References

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Figures

Fig. 0 — Two AlphaGo Encounters as Microcosm. Paired per-move KataGo policy rank and winrate trajectories. Left: Lee Sedol vs AlphaGo, Game 4, 2016. Right: Ke Jie vs AlphaGo Master, Game 1, 2017.

Source trace: results/figures/fig0_paired_vignette_trace.csv. Per-move rank and winrate data are available in the trace file. Rhythm effect sizes quoted in §1.1 were computed from per-game feature analysis (source data available upon request).

Fig. 1 — Century-scale structural drift envelope. (A) Decade-to-decade drift timeline with AI-era marked. (B) AI-era drift position relative to the pre-2016 drift distribution.

File: results/figures/fig1_drift_envelope.png

Fig. 2 — Fixed-horizon adoption probability by era of first appearance. Probability of reaching ≥ 3 unique players within 1, 2, 3, and 5 years.

File: results/figures/fig2_fixed_horizon_adoption.png

Fig. 3 — Two revolutions, two mechanisms. (A) 1910s→1920s drift decomposition into overlap vs non-overlap players. (B) Structural drift vs adoption speed across institutional and oracle revolutions.

File: results/figures/fig3_1910s_decomposition.png

Fig. 4 — Regional event-stream three-phase distance. Five-feature Euclidean distance across JPN-coded, KOR-coded, and CHN-coded streams with bootstrap 95% CIs.

File: results/figures/fig4_regional_three_phase.png

Fig. 5 — Pattern adoption lag and reuse density by era of first appearance.

File: results/figures/fig5_pattern_adoption.png

Supplementary Information

S1 — KataGo feature-level selective convergence

KataGo analysis across 200 positions (50 per era). At the aggregate level, the human-KataGo rhythm gap did not change across eras (Cohen's d , E0 vs E3 = -0.008). Feature-level analysis revealed selective convergence: E3 human moves were closer to KataGo policy than E0 human moves on `adj_opp` (E3 Cohen's d = -0.016 vs E0 d = -0.220), `adj_own` (E3 d = $+0.007$ vs E0 d = -0.209), and `dens_delta` (E3 d = -0.043 vs E0 d = $+0.216$). The other two dimensions (`dist_last`, `is_corner_open`) showed divergence or no change. KataGo model: `katal-b28c512nbt-s13255194368-d5935380940`, `maxVisits` = 100, Japanese rules, komi 6.5.

Fig. S1 — Feature-level human-KataGo rhythm gap by era. Positive Cohen's d = human closer to KataGo. (File: `results/figures/figS1_katago_alignment.png`; Data: `results/phase2/katago_alignment.csv`)

S2 — Prefix-length robustness for adoption lag

Adoption lag acceleration replicated at prefix lengths 10, 15, and 20 moves. Prefix 10 (N = 1,877 qualifying): 9.0→1.3 yr. Prefix 15 (N = 546 qualifying): 6.3→1.5 yr. Prefix 20 (N = 177 qualifying): 4.1→1.4 yr. The post-2016 additional reduction is strongest for prefixes 10 and 15; at prefix 20, both 2000–2015 and 2016+ show 1.4-year observed lag, implying that the AI-specific increment is undetectable at this coarser pattern granularity.

Fig. S2 — Adoption lag by era for prefix lengths 10, 15, and 20. (File: `results/figures/figS2_prefix_robustness.png`; Data: `phase5_defense.py` output)

S3 — Event-stream top-player leave-one-out control

The Korea-coded event-stream fighting index (`adj_opp` – `adj_own`, 2016+) was tested under leave-one-out removal of each of the top 10 players by game count in that stream. The index remained stable: the maximum shift upon removing any single player was $\Delta < 0.001$. The signal is distributed across the event-stream player pool and is not driven by any single individual. Note that event-stream labels reflect tournament provenance; players appearing in Korea-coded events include non-Korean-nationality participants (e.g., Ke Jie, Nakamura Sumire, Cho Chikun).

Fig. S3 — Korea-coded stream fighting index under per-player removal. (File: `results/figures/figS3_player_control.png`; Data: `phase5_defense.py` output, Section 2)